

PROJECT – CASE STUDY:

Optimizing Cryoprecipitate Collection for the American Red Cross:
<https://www.informs.org/Impact/O.R.-Analytics-Success-Stories/Saving-Lives-with-Analytics>

1. Execute Summary & Implementation Method:

Unlike standard whole blood, blood intended for cryoprecipitate—a critical clotting agent—must be transported from the mobile blood drive to the processing facility within an absolute maximum of 8 hours. This creates a massive logistical hurdle due to unpredictable donor turnouts and the geographic spread of mobile collection sites.

Overall System Motivation:

- **Given:** {The inherent unpredictability of human donor turnout, a geographically scattered network of temporary mobile blood drives, and an uncompromising 8-hour biological constraint for cryoprecipitate viability}
- **Use:** {An integrated, three-tiered analytics pipeline combining Predictive Forecasting, Stochastic Strategic Planning, and Real-Time Prescriptive Routing}.
- **To:** {Transform a reactive, high-waste supply chain into a precision-timed logistics network that maximizes life-saving blood production, minimizes driver operational costs, and practically eliminates the spoilage of expired blood}.

2. The Analytical Framework:

Model 1: Donor Yield Forecasting (Predictive):

The Red Cross cannot plan routing if they don't know how much blood to expect. Because donor behavior is highly variable (people get sick, it rains, people walk-in without appointments), this model establishes the "Expected Value" and the "Variance" of the blood supply at every single high school, church, and community center hosting a drive that day.

- **Motivation:**
 - **Given:** {Historical mobile drive collection records, day-of-week trends, weather forecasts, and the number of pre-registered donors for upcoming drives}.
 - **Use:** {Predictive Analytics, specifically Poisson or Negative Binomial Regression}.
 - **To:** {Forecast the expected number of viable whole blood units that will be collected at each specific mobile site on a given day}.
- **Mathematical Notation:**

Because we are predicting "counts" (number of donors/units of blood), we model the probability of collecting units of blood at site i on day t using a Poisson distribution:

$$P(Y_{it} = y) = \frac{e^{-\lambda_{it}} \lambda_{it}^y}{y!}$$

To link our predictors to the expected yield (λ_{it}), we use the log-link function:

$$\ln(\lambda_{it}) = \beta_0 + \beta_1 X_{1,it} + \beta_2 X_{2,it} + \dots + \beta_k X_{k,it}$$

Where:

- λ_{it} : The expected number of viable blood units at site i .
- $X_{k,it}$: The predictive features (e.g., X_1 = number of pre-registered donors, X_2 = precipitation index).

- **Data Architecture & Variables:**

To feed the above model, the below predictors/variables are what I would propose to be used:

- **Historical Variables (Static):**

- Site_ID, Site_Type (e.g., High School, Corporate, Church), Site_ZipCode.
- Historical_Yield_Avg (Average units collected at this site historically).
- Drive_Duration (Hours the mobile site is open).

- **Dynamic/Real-Time Variables:**

- Pre_Registered_Count (Number of appointments booked 24 hours prior).
- Day_of_Week_Index (Categorical: 1-7).
- Weather_Precipitation_Forecast (Inches of rain/snow expected).
- Weather_Temperature (Extreme heat/cold impacts walk-ins).
- Local_Event_Flag (Binary: Is there a competing local event/holiday?).

- **Missing Data & Handling Strategies:**

When operating mobile blood drives in church basements, high school gyms, and rural community centers, data will inevitably be lost or delayed. The system must fail gracefully.

- **The Challenge (Missing Variables):** A new mobile site is added with zero historical data. Alternatively, the weather API drops out during the Sunday night batch prediction.
- **The Handling Strategy: *Cold-Start Imputation:** For new site, use K-Mean Clustering to find "look-alike" sites based on zip code demographics, site type (e.g. University), and population density. Assign the cluster's historical average to the new site.

- **Conservative Defaulting:** If weather data is missing, the model should default to “Clear/Normal”. We never want to artificially lower the prediction (and risk understaffing) based on an assume phantom storm.

Model 2: Dynamic Site Selection (Prescriptive/Stochastic):

Not every blood drive should be a “cryo” drive because sending rapid=response couriers everywhere is prohibitively expensive. This model acts as the master strategist, adjusting daily targets based on inventory. If Monday and Tuesday have high donor turnouts and the region is ahead of schedule, the model will output a relaxed strategy for Wednesday, saving money. If there was a blizzard on Monday causing a massive deficit, the model will aggressively target maximum sites on Wednesday to ensure hospitals don't run out of cryo.

- **Motivation:**

- **Given:** { The predicted blood yields from Model 1, the weekly regional hospital targets for cryoprecipitate, current banked inventory levels, and the base costs of dispatching couriers}.
- **Use:** {Stochastic Dynamic Programming (SDP) to handle decision-making under uncertainty}.
- **To:** {Dynamically select which specific mobile blood drives should be designated as "expedited cryo-collection sites" each morning, adjusting the strategy based on how much cryo was successfully banked earlier in the week}.

- **Mathematical Notation:**

We use the Bellman Equation to find the optimal policy over the week. Let t be the day of the week, and S_t be the current inventory of cryo relative to the weekly target:

$$V_t(S_t) = \min_{x_t \in X} \{C(x_t) + \mathbb{E}_{W_t}[V_{t+1}(S_t + Y(x_t, W_t))]\}$$

Where:

- $V_t(S_t)$: The optimal expected cost from day t to the end of the week.
- x_t : The decision vector (which sites to activate for expedited cryo pickup today).
- $C(x_t)$: The fixed logistical cost of sending couriers to those selected sites.
- W_t : The random variable representing donor turnout (the uncertainty).
- $Y(x_t, W_t)$: The actual yield of blood realized from the selected sites.

- **Data Architecture & Variables:**

To feed the above model, the below predictors/variables are what I would propose to be used:

- Weekly_Cryo_Target (Total units required by regional hospitals).
- Current_Cryo_Inventory (Units currently banked and viable).
- Hospital_Burn_Rate (Average daily units requested by hospitals).
- Site_Activation_Cost (Fixed cost to designate a site for expedited pickup).

- **Missing Data & Handling Strategies:**

When operating mobile blood drives in church basements, high school gyms, and rural community centers, data will inevitably be lost or delayed. The system must fail gracefully.

- **The Challenge (Missing Targets):** Regional hospitals fail to report their midnight inventory of cryo, meaning the "Current_Cryo_Inventory" variable is blank, throwing off the stochastic targets.
- **The Handling Strategy:**
 - **Pessimistic Safety Stock:** In healthcare logistics, missing data must trigger the worst-case scenario assumption. If a hospital's inventory is missing, the model must assume they are at their **Maximum Burn Rate** and have zero buffer. The algorithm will automatically aggressively target more sites for cryo-expedition to ensure no patient bleeds out due to an IT error.

Model 3: Courier Routing with Time Windows (Prescriptive/Optimization):

This is the logistical execution layer. Once blood is drawn from a donor's arm, an invisible 8-hour countdown timer begins. If the blood arrives at hour 8 and 1 minute, it cannot be used for cryo. This model solves the complex puzzle of determining whether a driver should wait at a high school for 10 more donors to finish, or leave immediately to secure the blood they already have before it expires.

- **Motivation:**

- **Given:** {The designated cryo-collection sites for the day (from Model 2), real-time traffic data, and a fleet of available couriers}.
- **Use:** {Mixed-Integer Linear Programming (MILP) specifically formulated as a Vehicle Routing Problem with Time Windows (VRPTW)}.
- **To:** {Determine the exact minute-by-minute dispatch schedule and route for each courier to ensure all collected blood arrives at the central processing depot before the 8-hour expiration clock runs out}.

- **Mathematical Notation:**

Minimize Transportation Cost: $\min \sum_{k \in K} \sum_{i \in N} \sum_{j \in N} C_{ij} x_{ijk}$ *Subject to the biological constraint (The 8-Hour Rule):*

For any site i where blood is drawn at time $T_{draw,i}$:

$$a_{depot,k} - T_{draw,i} \leq 8 \text{ hours} \quad \forall i \text{ serviced by courier } k$$

Where:

- x_{ijk} : Equals 1 if courier travels directly from site i to site j .
- C_{ij} : The travel cost/distance between nodes.
- $a_{depot,k}$: The time courier k arrives back at the processing facility.

- **Data Architecture & Variables:**

To feed the above model, the below predictors/variables are what I would propose to be used:

- **Spatial Variables:**

- Depot_Lat, Depot_Long (GPS of the central processing facility).
- Site_Lat, Site_Long (GPS of all active mobile drives).
- Distance_Matrix (Real-time driving distances between all active nodes).

- **Temporal Variables (The 8-Hour Clock):**

- Time_of_Draw (Exact timestamp when the needle is removed from the donor).
- Time_to_Process (Buffer time needed at the depot to process the blood once it arrives, e.g., 30 mins).
- Expiration_Timestamp (Time_of_Draw + 8 hours - Time_to_Process).

- **Fleet Variables:**

- Courier_ID, Vehicle_Capacity (Max coolers a van can hold).
- Courier_Shift_Start, Courier_Shift_End.
- Live_Traffic_Multiplier (Real-time delay coefficient from a traffic API).

- **Missing Data & Handling Strategies:**

When operating mobile blood drives in church basements, high school gyms, and rural community centers, data will inevitably be lost or delayed. The system must fail gracefully.

- **The Challenge (The Missing Timestamp):** A phlebotomist is overwhelmed with donors, the tablet loses Wi-Fi connection, and they cannot log the exact Time_of_Draw for a cooler of blood. The 8-hour countdown timer is now completely unknown.

- **The Handling Strategy:**

- **Left-Bound Bounding:** The model must assume the absolute earliest possible time that blood could have been drawn. If the drive started at 8:00 AM and the tablet syncs at 11:00 AM with missing timestamps, the model assigns the 8-hour clock starting at **8:00 AM**. This drastically shrinks the delivery window but guarantees the biological safety of the product.
- **Offline-First Routing Fallback:** If the live traffic API (like Waze/Google) drops, the solver must immediately switch to a static, locally hosted **Dijkstra's algorithm** distance matrix based on speed limits.

3. Data Collection Methods & Infrastructure to Consider:

To feed this three-tiered system, the RedCross probably needs a robust, multi-channel data architecture.

- **Mobile Edge Computing (The Point of Care):** Phlebotomists use ruggedized tablets equipped with barcode scanners. These apps must be built with "Store-and-Forward" architecture. If the rural high school has no cell service, the tablet stores the timestamped scans locally and pushes them to the cloud via a message broker (like Apache Kafka) the instant the van hits a cell tower.
- **Telematics & IoT:** Courier vehicles must be outfitted with dual-sensor IoT devices.
 - **GPS Tracking:** Pings lat/long coordinates every 10 seconds to update the routing solver.
 - **Cooler Telemetry:** Active temperature monitoring. If a cooler breaches the strict temperature threshold, the routing model instantly flags that blood as "spoiled" and reroutes the driver to save the rest, rather than rushing a ruined product.
- **External API Hooks:** Dedicated data pipelines pulling from the National Weather Service (NOAA) for 7-day forecasts and state Department of Transportation APIs for real-time road closures.

4. Data Refresh Strategies to Consider:

An analytics system is a living organism; different parts must "breathe" at different rates.

- **Long-Term Strategy (Monthly/Quarterly):** The Poisson Forecasting model (Model 1) undergoes a full retrain monthly. This captures seasonal macro-trends (e.g., high school students don't donate in the summer; flu season depresses winter donations).
- **Daily Batch Strategy (3:00 AM):** The Stochastic Site Selection model (Model 2) runs once per night. It executes a batch ETL (Extract, Transform, Load) job pulling the finalized midnight inventory from all regional hospitals to set the operational targets for the sunrise shift.
- **Micro-Batch / Streaming Strategy (Every 5 Minutes):** The VRPTW Routing model (Model 3) is a continuous stream. Using a framework like Apache Spark Streaming, it recalculates the remaining 8-hour time windows and driver distances every 5 minutes. If a driver hits gridlock, the system quietly recalculates and pushes a new route to their dashboard before they even realize they are delayed.

5. Legal and Ethical Considerations:

Operations Research in healthcare carries massive ethical weight. Efficiency cannot supersede human rights or equity.

A. HIPAA and Data Minimization (Legal):

A blood bag contains highly sensitive Protected Health Information (PHI), including infectious disease markers.

- **The Rule:** The routing algorithm and the courier do not need to know who the blood belongs to or what is in it.
- **The Solution:** Data masking. The optimization model only receives a cryptographic hash (e.g., Bag_ID: A8X9), its location, and its Time_of_Draw. All demographic and medical data is stripped at the tablet level before reaching the logistics server.

B. The Equity of Site Selection (Ethical):

Mathematical optimization is inherently ruthless. If left unchecked, the Site Selection model (Model 2) will realize that dispatching couriers to rural, minority, or lower-income neighborhoods is "statistically inefficient" due to longer drive times and historically lower yields.

- **The Danger:** If the algorithm simply stops selecting these sites for cryo-processing, it leads to a catastrophic lack of diversity in the blood supply (which is vital for treating conditions like Sickle Cell Disease, which requires ethnically matched blood).
- **The Solution:** The objective function must include an Equity Constraint. The model must mathematically be forced to select a minimum percentage of sites from diverse zip codes every week, even if it incurs a higher logistical cost.

C. Labor Laws and Courier Safety (Legal/Ethical):

An algorithm cannot prioritize the 8-hour biological clock over the safety of the human driver.

- **The Danger:** A delayed driver might be routed by the algorithm to travel 30 miles in 25 minutes to save a cooler of blood, inadvertently encouraging speeding or reckless driving.
- **The Solution:** The VRPTW constraints must strictly cap maximum travel speeds at the legal limit. Furthermore, the model must hard-code mandatory union or state-mandated rest breaks for the drivers, making those time blocks "unavailable" for routing, regardless of the blood's expiration timer.

D. The “Trolley Problem” of Spoilage (Ethical):

If a massive traffic jam occurs and a courier can only reach Site A (which has 5 units of blood) or Site B (which has 5 units of blood) before the 8-hour clock expires, how does the machine choose?

- **The Solution:** This cannot be a black-box decision. The Red Cross must program a deterministic hierarchy. For example, if Site A collected blood from a rare O-negative donor, the system must prioritize Site A. If all things are equal, the algorithm should default to whichever site minimizes the driver's risk.

6. Proposed Final System Integration:

The Integrated Flow:

1. **Sunday Night (Planning Phase): Model 1** runs to generate expected donor yields for every scheduled mobile drive for the upcoming week.
2. **Daily at 5:00 AM (Strategic Phase): Model 2** evaluates the current regional inventory against the weekly hospital demand. It looks at Model 1's predictions and selects the specific subset of sites that will be "cryo-designated" for that day.
3. **Intra-Day (Execution Phase): Model 3** runs continuously throughout the day. As site supervisors input actual blood collection numbers into their tablets, the MILP solver adjusts courier routes dynamically, sending vans to intercept blood well before the 8-hour limit.

Cost of Error Constraints: The analytics component here must heavily penalize the **False Negative** in routing (assuming a driver will make it in 8 hours, but they take 8 hours and 5 minutes due to traffic). To mitigate this, the constraints in Model 3 should technically be set to an artificial threshold of **~7.25 hours** to create a safety buffer for processing intake at the depot.

7. Validation Challenges & Pre-Deployment Testing:

Challenges

- **Data Latency:** Rural cell service dead zones delaying the start of the 8-hour clock.
- **Lumpy Arrivals:** Predicting smooth Poisson curves vs. a sudden influx of 40 donors at once, maxing out van capacity.
- **Traffic Volatility:** Deterministic routing models failing when a sudden accident causes gridlock.

Testing Protocols:

- **Historical Backtesting:** Run the models on prior year data to see if they theoretically would have collected more cryo than the legacy system.
- **Monte Carlo Stress Testing:** Simulate 10,000 days injecting extreme variables (blizzards, viral donor surges) to ensure constraints hold.
- **Shadow Deployment:** Run the algorithm live alongside human dispatchers without letting it control drivers to compare routing efficiency.
- **Geographic A/B Piloting:** Roll the system out in one state while a similar state serves as the control group for 90 days.

8. Key Performance Indicators (KPIs) :

1. **Spoilage Rate (Primary):** *(Units exceeding 8-hour limit / Total units collected).*
Goal: < 2%.

2. **Cryo Yield Fulfillment (Primary):** *(Total viable units processed / Weekly regional target)*. Goal: 100%.
3. **Logistical Cost per Unit (Secondary):** *(Daily courier wages + fuel / Total viable units)*. Goal: Consistent cost reduction.
4. **Courier Utilization Rate (Secondary):** *(Hours transporting blood / Total shift hours)*. Goal: Maximum active utilization.